```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is cPIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=3)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=3)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=3)
)
Total trainable parameters: 480
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:09<00:00, 217.09epoch/s, Loss=5.
00e-06, LR=1.0e-03, Best=5.00e-06]
Training finished. Best loss: 3.85e-06. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 10.000862836837769s
Relative loss: 0.05607824772596359
L2 loss: 0.035682667046785355
Parameter containing:
tensor([[[ 0.0771,  0.1370,  0.4731, -0.0758],
         [-0.0302, -0.5360, -0.2149,  0.1844],
         [-0.0762,  0.3004,  0.3420,  0.0293],
         [-0.1129, -0.2052, -0.0790,  0.7767],
         [ 0.1902, -0.1470, -0.5414, -0.0962],
         [-0.6142,  0.1375,  0.3459, -0.0540],
         [ 0.2846, -0.0107, -0.2681, -0.2348],
         [ 0.0053,  0.3208, -0.3298, -0.3085],
         [ 0.6692, -0.1669, -0.6302, -0.3605],
         [ 0.3061, -0.3243, -0.5708, -0.0897]]], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[[-6.6935e-03,  2.2485e-01,  1.3564e-02, -1.9712e-01],
         [ 4.9177e-02, -2.2349e-01, -6.7673e-02,  2.4096e-01],
         [ 2.1989e-02,  1.2314e-01, -1.7651e-02, -8.3625e-02],
         [ 1.0609e-02, -2.2145e-01, -2.6230e-02,  2.0471e-01],
         [-2.8673e-02,  2.1855e-01,  4.9380e-02, -1.5361e-01],
         [ 3.8705e-02, -2.0009e-01, -3.3478e-02,  1.9860e-01],
         [ 2.1784e-02, -1.8321e-01, -2.0732e-02,  1.6662e-01],
         [ 4.6542e-02, -2.1426e-01, -8.0545e-03,  1.5211e-01],
         [-2.3182e-02,  1.9521e-01, -1.6228e-02, -2.4335e-01],
         [-2.8407e-02,  2.4115e-01, -8.0032e-03, -1.7404e-01]],

        [[-4.3503e-02, -2.1345e-01,  5.0718e-02,  1.8644e-01],
         [ 3.3725e-02,  2.0647e-01, -8.9517e-03, -1.5688e-01],
         [ 3.6961e-02, -2.1763e-01, -9.1694e-03,  2.3961e-01],
         [ 4.6147e-02,  2.0489e-01, -4.6656e-02, -2.3165e-01],
         [-2.2892e-02, -2.0426e-01,  4.1809e-02,  1.7099e-01],
         [-1.6553e-02,  1.6250e-01, -6.0842e-02, -1.6902e-01],
         [ 4.7894e-02,  2.2649e-01,  4.7636e-04, -1.7714e-01],
         [-2.3061e-02,  2.1288e-01, -1.7219e-02, -2.0593e-01],
         [-5.3193e-03, -2.8073e-01, -5.3022e-03,  2.3143e-01],
         [-5.9988e-02, -2.0310e-01,  5.9604e-02,  1.7851e-01]],
```

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       [[-2.1016e-02,  1.7024e-01, -2.9609e-02, -1.8621e-01],
        [ 8.4761e-02, -2.9388e-01, -6.1615e-02,  2.8315e-01],
        [ 2.4444e-03,  8.9654e-02,  6.0377e-03, -2.4007e-02],
        [ 2.0961e-02, -2.3181e-01,  7.4950e-03,  2.0816e-01],
        [-8.4744e-02,  2.6039e-01,  6.1597e-02, -2.1171e-01],
        [ 5.1524e-02, -1.5700e-01, -3.1275e-02,  2.1372e-01],
        [ 2.7178e-02, -2.1475e-01, -1.8577e-02,  1.6268e-01],
        [ 3.0631e-02, -2.0204e-01, -3.2052e-02,  1.4303e-01],
        [-4.6335e-02,  2.2586e-01, -3.8330e-02, -2.1868e-01],
        [-6.2172e-03,  1.9394e-01,  1.1567e-02, -2.0792e-01]],

       [[-4.7082e-02, -7.1323e-02, -2.9709e-02, -5.3677e-03],
        [ 9.4838e-02, -6.2325e-02,  2.8150e-02, -8.0556e-02],
        [ 8.5428e-02, -1.4291e-01,  2.7445e-02,  1.1102e-01],
        [-4.3773e-03,  1.0437e-01,  1.4329e-02, -8.4628e-02],
        [-4.5188e-02, -1.4667e-03,  1.1479e-03,  2.8941e-02],
        [ 3.5935e-02, -1.2140e-02, -2.3756e-02, -2.5497e-02],
        [-8.2899e-04, -1.7375e-03, -3.7597e-02, -2.0763e-02],
        [ 2.1263e-02,  1.4058e-01, -5.5149e-02, -1.8303e-01],
        [-1.9385e-02, -8.4421e-03,  4.2593e-02,  1.6034e-02],
        [-1.1923e-02, -3.7232e-02,  5.4620e-02,  4.5152e-02]],

       [[-1.9112e-02, -6.7843e-02, -4.7003e-03,  3.9020e-02],
        [-5.7432e-03,  1.4637e-01, -2.0508e-02, -1.3918e-01],
        [ 4.3191e-03, -5.4420e-03, -2.9766e-02, -4.1393e-02],
        [ 6.2274e-03,  1.5262e-02, -8.3081e-03, -3.9573e-02],
        [-3.9018e-02, -6.5533e-02,  2.7758e-03,  8.5477e-02],
        [ 7.5042e-03,  1.2035e-01, -2.1959e-02, -3.1118e-02],
        [ 1.3166e-02,  3.6763e-02, -2.0834e-02, -9.0690e-02],
        [ 3.0073e-03,  7.7343e-02, -4.1924e-02, -5.7697e-02],
        [ 2.6016e-03, -7.6047e-02, -1.0333e-02, -8.0379e-03],
        [-2.6956e-02, -6.9638e-02, -1.6318e-02,  8.5418e-02]],

       [[-3.1553e-02,  4.3613e-02, -2.1025e-02, -2.0624e-02],
        [ 4.6117e-02, -2.9233e-02,  6.5440e-02,  6.6538e-02],
        [ 4.8817e-03, -1.5078e-02, -5.3528e-02,  6.5103e-03],
        [ 1.2130e-02, -8.3232e-02,  7.0481e-02, -2.3400e-02],
        [-2.4438e-02,  1.6961e-02, -2.7943e-02, -6.5484e-02],
        [ 4.9951e-02, -2.2421e-03,  5.5824e-02,  3.6716e-02],
        [ 2.9654e-02, -3.5667e-02,  1.0408e-02, -9.9790e-03],
        [ 1.5411e-02, -5.1203e-02,  1.0143e-01, -2.6482e-02],
        [-5.5934e-03,  1.7047e-02, -6.5055e-03, -1.7984e-02],
        [-4.7132e-02,  8.4454e-03, -4.1714e-02, -1.8044e-02]],

       [[ 3.0273e-02, -1.0755e-02,  7.1506e-02,  3.7967e-02],
        [ 3.3588e-02,  5.6867e-02,  3.3393e-02, -1.0129e-01],
        [-6.0171e-02,  3.7063e-02,  2.5753e-02,  4.8018e-03],
        [-2.0006e-02,  1.1496e-02, -2.9065e-02, -1.5093e-02],
        [-1.0590e-02, -3.0914e-02,  3.1536e-03, -9.3211e-03],
        [ 7.1536e-03,  2.8131e-02,  5.6253e-04, -2.5776e-02],
        [ 5.8773e-02,  3.5622e-02, -4.4595e-04, -5.1414e-02],
        [ 4.0375e-02, -9.0134e-03, -8.1142e-02, -3.9226e-02],
```

```
          [ 6.1361e-03, -6.5943e-03,  5.0933e-02, -1.7467e-03],
          [-6.4495e-02, -3.5279e-02,  1.6701e-02,  1.1197e-02]],

         [[-9.7846e-02, -2.4699e-02,  2.4312e-02,  4.8456e-02],
          [ 5.1110e-02,  1.0564e-01, -2.9682e-03, -3.0446e-02],
          [-1.0650e-02,  2.1472e-02,  4.2533e-02, -7.2121e-02],
          [ 7.2148e-02, -3.9747e-02, -6.6219e-02,  1.6024e-02],
          [-4.4570e-02,  4.9036e-02,  5.2784e-02,  6.1034e-02],
          [ 2.3102e-02,  2.9084e-02, -1.3594e-02, -1.7503e-02],
          [-9.1732e-03,  5.0578e-02, -2.7648e-02, -1.3923e-02],
          [ 3.8090e-02, -3.5462e-02, -5.8323e-02,  2.9078e-02],
          [-5.0691e-02,  2.9459e-03,  4.2476e-02,  1.6021e-02],
          [-4.9029e-02, -4.0591e-02,  4.7915e-03, -7.1236e-04]],

         [[ 1.8944e-02, -3.0152e-02, -9.8425e-02, -1.0730e-02],
          [ 4.5778e-02,  9.3749e-02,  2.7881e-01, -8.1690e-03],
          [-2.7196e-03,  5.6793e-02,  4.9077e-02,  1.2500e-02],
          [-5.3989e-03,  8.1058e-03,  7.3656e-02, -5.1302e-02],
          [-3.0167e-02, -4.9797e-02, -1.8151e-01,  5.5780e-03],
          [-1.3899e-02,  3.2499e-02,  1.1665e-01, -3.0690e-02],
          [ 4.9379e-02,  2.0385e-03,  1.5220e-01, -1.5112e-02],
          [ 3.7548e-02,  2.8495e-02,  5.5472e-02, -8.7818e-02],
          [ 3.0146e-02, -1.6360e-02, -6.3666e-02,  3.5999e-02],
          [-2.1057e-02, -5.3767e-02, -1.2173e-01,  5.5638e-02]],

         [[-9.6862e-03, -2.8149e-02,  4.4495e-02,  6.7817e-02],
          [ 3.4762e-02,  1.2572e-01,  1.1597e-02, -1.5274e-01],
          [ 3.0895e-02,  2.1612e-02, -2.2332e-02, -1.7720e-02],
          [ 1.5022e-02,  7.1498e-02,  4.0855e-02, -3.9052e-02],
          [-2.6319e-02, -5.7630e-02, -1.6628e-02,  5.6822e-02],
          [ 4.2144e-02,  5.7558e-02,  2.3464e-05, -6.3512e-02],
          [ 5.5216e-02,  4.4614e-02, -1.4771e-02, -2.8737e-02],
          [ 3.0424e-02,  8.4647e-02,  1.3201e-01, -2.3893e-02],
          [-8.6626e-02, -5.2717e-02, -4.3046e-02,  3.1900e-02],
          [-1.8939e-02, -9.0698e-02, -1.2607e-02,  4.8963e-02]]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[-6.1784e-03, -1.5823e-02, -2.9125e-02,  1.3766e-01]],

         [[ 2.7283e-03,  2.6470e-02, -2.4984e-02, -6.8188e-02]],

         [[ 3.2144e-02,  4.9455e-02,  2.0406e-02, -9.3907e-02]],

         [[ 6.4230e-02,  1.2027e-02, -8.0601e-02, -7.8529e-02]],

         [[-1.6423e-02, -2.2444e-03, -1.1089e-04,  7.0063e-02]],

         [[ 3.3324e-02,  1.6229e-02, -2.1935e-02, -1.0030e-01]],

         [[ 5.8328e-03,  1.6220e-03,  1.6244e-02, -7.3220e-02]],

         [[ 6.9459e-03, -1.7190e-02, -4.5917e-02, -9.0029e-02]],

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        [[ 5.3231e-02, -5.6573e-04, -5.0811e-02,  5.8052e-02]],

        [[-2.8235e-02, -2.3613e-02, -1.6528e-02,  5.1898e-02]]],
       device='cuda:0', requires_grad=True)
==========================================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================================
CPIKANModel                              [1, 1]                    --
├─ModuleList: 1-1                          --                        --
│    └─0.weight                                                        ├─40
│    └─1.weight                                                        ├─400
│    └─2.weight                                                        └─40
│    └─ChebyKANLayer: 2-1                  [1, 10]                   40
│    │    └─weight                                                      └─40
│    └─ChebyKANLayer: 2-2                  [1, 10]                   400
│    │    └─weight                                                      └─400
│    └─ChebyKANLayer: 2-3                  [1, 1]                    40
│    │    └─weight                                                      └─40
==========================================================================================
Total params: 480
Trainable params: 480
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==========================================================================================
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==========================================================================================

进程已结束，退出代码为 0

```